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Rediscovering the Standard Model with AI

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Abstract We investigate whether artificial intelligence can reproduce and organize known structures of the Standard Model of particle physics using experimental data and with minimal theoretical inputs. By applying unsupervised machine learning techniques — including data dimensionality reduction and clustering algorithms — to intrinsic particle properties and decay modes, we show that key organizational features of particle physics, such as the relative strength of different interactions and the difference between baryons and mesons, become evident in the resulting data representations. We also identify conserved quantities such as baryon number, strangeness and charm as well as the structure of isospin and the Eightfold Way multiplets. Our analysis then reveals that clustering can separate particles by interaction, flavor symmetries as well as quantum numbers. Additionally, we observe patterns consistent with Regge trajectories in baryon excitations. Our results show that machine learning can highlight known key structures of the Standard Model from data, suggesting a promising path toward data-supported discovery in fundamental physics.

Keywords Flavour Symmetries, Bottom Quarks

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Introduction

Artificial intelligence (AI) is rapidly establishing itself as a powerful tool across a wide range of scientific disciplines, thanks to its ability to deliver insights and accelerate research. In recent years, machine learning (ML) in particular has found increasing applications in various areas of physics — from experimental high-energy physics to more foundational domains (see for instance [1 6]). This growing relevance raises a natural and important question: can AI contribute to the formulation of fundamental physical theories?

Before addressing such an ambitious goal, it is essential to test whether AI systems are capable of identifying known physical structures from empirical data, without any built-in theoretical input.[1] In other scientific fields, encouraging progress has already been made in this direction. In chemistry, for example, unsupervised machine learning techniques have been successfully applied to rediscover the periodic table of elements from data on atomic environments and chemical compounds [7].

This paper explores whether a similar strategy can be applied to the domain of particle physics. Specifically, we investigate whether unsupervised machine learning techniques can reflect the internal structure and symmetries of the Standard Model (SM) — currently the most complete theoretical framework describing the known elementary particles and three of the four fundamental forces — in non-trivial representations learned from experimentally measured data..[2] Even though our work aims to identify the global symmetries of the Standard Model by just determining conserved charges like baryon number or isospin or flavor

[1] Note however that, even though our analysis itself will not use explicit theoretical model predictions or quantities, the input observables rely on theoretical assumptions. To be fair we may say that the machine learning algorithms are not actually discovering a latent structure but they build up on previously established theoretical constructs.

[2] See [8–14] for different AI approaches to particle physics phenomenology. In particular, in [8] a study is performed obtaining properties of baryons from that of mesons and, unlike the present paper, assuming they are composed by quarks.

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SU(3) multiplets, we do not use nor attempt to derive these symmetries. In some sense our work is orthogonal to recent approaches to describe symmetries from machine learning [15 17].

The goal of this work is to assess whether the key organizational features of the Standard Model can be captured from experimental data when analyzed by unsupervised representation methods.. To this end, we divide the paper into two parts. In section 2, we introduce the unsupervised learning techniques most relevant for this study, including principal component analysis, t-distributed stochastic neighbor embedding, and clustering algorithms. In section 3, we apply these tools to real particle physics data — focusing first on decay modes and then separately on intrinsic particle properties — to examine whether meaningful structures resembling those of the Standard Model can be recovered.

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2 Data analysis techniques

2.1 Data dimensionality reduction

In many real-world datasets, especially in particle physics, the number of variables used to characterize each data point can be large, making it difficult to visualize, interpret, or extract meaningful patterns. Dimensionality reduction refers to a set of techniques that transform high-dimensional data into a lower-dimensional representation while preserving as much relevant structure as possible. This process facilitates both data visualization and downstream analysis by highlighting key relationships and reducing noise or redundancy.

In this section, we focus on two complementary approaches to dimensionality reduction. First, we examine principal component analysis, a linear technique that projects the data onto orthogonal directions of maximal variance. Then, we turn to t-distributed stochastic neighbor embedding, a nonlinear method designed to preserve local neighborhood structures in a reduced space. For more details on these methods, refer to [18 22].

Principal Component Analysis. Principal Component Analysis (PCA) is a statistical method for dimensionality reduction based on linear transformation. Given a dataset $X \in {}^{n,m}$ with n samples and m features, PCA transforms the original correlated variables into a new set of uncorrelated variables, known as principal components. These components are constructed as linear combinations of the original features in such a way that they correspond to the eigenvectors of the covariance matrix of the data. Let $\bar{X}$ be the centered scaled dataset,[3] the covariance matrix is then given by

$$C = \frac{1}{n \quad 1} \bar{X}^T \bar{X} \ . \tag{2.1}$$

The covariance eigenvectors v_i are obtained by solving the eigenvalue problem

$$C v_i = \lambda_i v_i \tag{2.2}$$

where the eigenvalues λ_i indicate the amount of variance captured by each corresponding component.

[3]Here we refer to the standard scaled dataset, i.e., $X_{ij} = \frac{X \ j \quad j}{\sigma_j}$, where $\ _j$ and σ_j are the mean and standard deviation of feature j. The normalisation ensures that there is no bias due to different feature scales.

The principal components are typically ordered by decreasing eigenvalue, so that the first principal component captures the largest amount of variance in the dataset and each subsequent component capturing the maximum remaining variance under the constraint of orthogonality to the other ones. The dimensionality reduction is then achieved by retaining only the first k components, which are sufficient to preserve most of the original variance, and by projecting the data onto the corresponding lower-dimensional subspace:

$$X'_k = \bar{X} V_k \tag{2.3}$$

where V_k is the $m \times k$ matrix constructed with the first k eigenvectors. In this way, PCA reduces the complexity of the dataset while preserving its most significant variance structure. This facilitates the identification of dominant patterns and correlations within the data, allowing for more effective analysis and interpretation.

It is worth noting that PCA is closely related to the singular value decomposition (SVD) procedure used in [7]. In the case of SVD, the centered data matrix is factorized as

$$\bar{X} = U\ \Sigma\ V \tag{2.4}$$

where U and V are, respectively, $n \times n$ and $m \times m$ orthogonal matrices, and $\Sigma = \mathrm{diag}(\sigma_i)$ is a $n \times m$ rectangular diagonal matrix whose entries σ_i are the singular values of $\bar{X}$. Since $\bar{X}^T \bar{X} = V^T \Sigma^T U^T U \Sigma V = V^T \Sigma^T \Sigma V$, the covariance matrix in (2.2) is written as

$$C \equiv \frac{1}{n \quad 1}\ V^T \Sigma^2\ V\ . \tag{2.5}$$

The covariance eigenvalues are related to the singular values by $\lambda_i = \frac{{}^2_i}{n\ \ 1}$, and the dimensionality reduced dataset is

$$X'_k = U_k\ \Sigma_k \tag{2.6}$$

where U_k and Σ_k are obtained considering the first k columns.

t-distributed stochastic neighbor embedding. t-Distributed Stochastic Neighbor Embedding (t-SNE) is a nonlinear manifold learning algorithm commonly used for the visualization of high-dimensional datasets. It is specifically designed to capture and preserve local structure in the data, meaning it focuses on maintaining the relative similarity of nearby data points. In contrast to linear dimensionality reduction techniques, which preserve global variance across all features, t-SNE emphasizes the accurate representation of local neighborhoods.

The algorithm constructs a probability distribution over pairs of high-dimensional objects $x_i, x_j \in X$, such that similar objects (close neighborhoods) have a higher probability of being selected. This is typically done by modeling pairwise similarities as Gaussian-distributed conditional probabilities centered at each data point

$$p_{j|i} = \frac{\exp\left(\quad ||x_i \quad x_j||^2 / 2\sigma_i^2 \right)}{\sum_{k \neq i} \exp\left(\quad ||x_i \quad x_k||^2 / 2\sigma_i^2 \right)}\ . \tag{2.7}$$

The parameters σ_i control the effective number of nearest neighbors for each point x_i and are determined by fixing the *perplexity*, defined as $\xi_i \equiv 2^{H_i}$ in terms of Shannon entropy

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for the conditional distribution, $H_i \equiv \quad \sum_j p_{j|i} \log_2 p_{j|i}$. In this context, the perplexity acts as a hyperparameter of the algorithm and sets the scale over which the local structure is preserved.[4] Given the conditional probability $p_{j|i}$, the corresponding symmetric joint distribution is obtained as $p_{ij} = \frac{p_{j|i} \quad p_{i|j}}{2n}$, with n number of data points.

In the low-dimensional embedding, a corresponding joint distribution q_{ij} is defined using a Student t-distribution with one degree of freedom,[5]

$$q_{ij} = \frac{(1 + ||y_i \quad y_j||^2)^{\quad 1}}{\sum_{k \neq l}(1 + ||y_k \quad y_l||^2)^{\quad 1}} \,. \tag{2.8}$$

The goal is to make q_{ij} as similar as possible to p_{ij} in order to preserve the information on the structure of the dataset. This is achieved by minimizing the Kullback-Leibler (KL) divergence between the two distributions:

$$D_{KL}(p|q) \equiv \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \,. \tag{2.9}$$

The divergence acts as a loss function, and the low-dimensional coordinate y_i in (2.8) are iteratively updated via a gradient descent method applied on $D_{KL}(p|q)$. Data points that are close together in high-dimensional space remain close in the embedding, while dissimilar points are modeled to be farther apart.

We note that the choice of a Gaussian and Student-t distributions is standard and, in general, other distributions could be adopted provided they serve the purpose of modeling similarities. Different choices primarily affect the global arrangement of the points while the local neighborhood structures, and hence the resulting groupings of interest, are expected to be more robust.

2.2 Clustering methods

Clustering is an unsupervised machine learning approach aimed at partitioning a dataset into distinct groups (clusters) such that data points within the same group are more similar to each other than to those in different groups. Unlike dimensionality reduction techniques, which project data onto a new coordinate system, clustering algorithms operate directly in the feature space, attempting to uncover the intrinsic grouping structure of the data without prior labeling.

-means clustering. K-means is one of the most widely used partition-based clustering algorithms. It operates by defining K centroids (cluster centers) and assigning each data

[4] Consistently with standard practice, a single global perplexity (the same for each i) is chosen such that $\xi_i < N$ and sufficiently small to emphasize local properties of the data distributions. While quantitative details of the embedding can depend on the specific value chosen for the perplexity, the qualitative structures presented in this work are stable under moderate variations of ξ.

[5] The use of a Student t-distribution in the low-dimensional space addresses the tendency of dimensionality reduction to compress large pairwise distances. Unlike a Gaussian, the t-distribution has heavier tails, which allow dissimilar points — those far apart in the high-dimensional space — to remain well-separated in the low-dimensional embedding.

point to the nearest centroid, thereby forming K clusters. The algorithm seeks to minimize the within-cluster sum of squares (WCSS):

$$D_{WCSS} \equiv \sum_{i=1}^{K} \sum_{x \in C_i} ||x - \mu_i||^2 \tag{2.10}$$

where C_i is the set of points in the i-th cluster and μ_i is the centroid of the cluster. The algorithm follows an iterative refinement approach:

- Choose K initial centroids.
- Assign each data point to the nearest centroid based on Euclidean distance.
- Recompute the centroids as the mean of the points assigned to each cluster.
- Repeat the previous two steps until convergence.

K-means assumes spherical, equally sized clusters and may perform poorly on data with non-convex shapes or varying density. It is sensitive to initialization and requires the number of clusters K to be specified in advance.

To quantitatively assess the quality of clustering, one can introduce specific evaluation metrics. In particular, for a given point i one can evaluate the balance between the mean intra-cluster distance (i.e., the average distance to points in the same cluster), $D_{ICi} \equiv \frac{1}{|C_i|-1} \sum_{j \in C_i, j \neq i} D_{ij}$, and the smallest mean distance to the nearest neighboring cluster, $D_{\mathrm{SNC}i} \equiv \min_{C \neq C_i} \left(\frac{1}{|C|} \sum_{j \in C} D_{ij} \right)$, by means of the Silhouette score defined as

$$S_s = \frac{1}{n} \sum_i S_{si} \equiv \frac{1}{n} \sum_i \frac{D_{SNCi} - D_{ICi}}{\max(D_{ICi}, D_{SNCi})} \,. \tag{2.11}$$

The Silhouette score ranges from -1 to 1: scores $S_s \approx 1$ indicate that the samples are well clustered in terms of cluster separation; values of $S_s \approx 0$ indicate overlapping clusters; negative scores suggest that the points are not properly clustered.

Hierarchical clustering. Hierarchical clustering is a more flexible technique that produces a nested hierarchy of clusters organized in a tree-like structure (dendrogram). It does not require the number of clusters to be specified a priori, and clusters can be obtained by cutting the dendrogram at a chosen level of granularity. In the next section, we will make use of the agglomerative approach which starts with each data point as its own cluster and recursively merges the closest pairs of clusters. The closeness between clusters will be defined with the Ward method which consists into minimizing the total within-cluster variance defined in eq. (2.10). The algorithm consists in the following steps:

- Start with n clusters defined by each point $x_i \in X$.
- Compute the within-cluster variance.
- Merge the pair of clusters such that the increase in the WCSS is minimized.
- Update the within-cluster variance.
- Repeat the previous two steps until only a single cluster remains.

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The iteration results can be visualized via the corresponding dendrogram, where the vertical height at which clusters are merged reflects the dissimilarity between them. By selecting a cut-off height, one can extract a flat clustering with a desired number of clusters or based on distance thresholds. With respect to K-means, hierarchical clustering has the advantages of being more informative and interpretable, as it captures multiscale structures and does not require a fixed number of clusters.

Again, clustering results depend on the choice of distance metric. While alternative metrics could in principle be considered, different choices primarily affect quantitative details of the cluster boundaries, whereas the existence of well-separated groups and the qualitative clustering patterns discussed here are expected to be robust under variations of reasonable metrics.

3 Applications in particle physics

In this section, we apply PCA and t-SNE as dimensionality reduction techniques to study patterns in both particle decay channels and intrinsic physical properties. Each particle can be described by a combination of these features, or by the set of possible decay products, forming high-dimensional representations. By reducing these spaces to two or three dimensions, and by applying clustering methods, we aim to uncover underlying structures and groupings that may reflect fundamental particle families, symmetries, and interaction mechanisms. The data used for this analysis are taken from the Particle Data Group (PDG) database [23].

3.1 Interactions from decay rates

Before delving into machine learning results, it is useful to reflect on how much physical insight can be obtained simply by examining basic observables. A clear example is the distribution of Standard Model particles in the mass-lifetime plane [24] shown in figure 1, which includes both hadrons and elementary particles. The most immediate observation from this plot is the presence of distinct clusters. One group, found at relatively short lifetimes (order $10^{\ 22}$ s), includes particles that decay predominantly via the strong interaction. These are mainly hadrons whose decay is driven by the large coupling strength of the strong force. The strong interaction leads to extremely rapid decay processes, consistent with the short lifetimes observed. Even though quarks and gluons are never seen in isolation due to confinement, their presence is indirectly encoded in the decay properties of hadrons. In contrast, a second, more diffuse cluster appears at longer lifetimes, ranging from $10^{\ 12}$ to $10^{\ 6}$ seconds. These particles decay primarily through the weak interaction, which is several orders of magnitude weaker than the strong force. As a result, their decay processes are rarer and occur on longer timescales. This group includes charged leptons, neutrinos, and weakly decaying hadrons. The pattern is particularly striking because it emerges without any explicit labeling by interaction type; it arises naturally from physical decay data.

This simple plot therefore serves as a compelling demonstration of how particle properties reflect underlying interaction dynamics. Particles subject to strong decays cluster separately from those governed by weak decays, solely on the basis of lifetime.

Figure 2 shows the same plot limited to the particles known before 1960. It is worth noting that π_0, η mesons and Σ^0 appear between the two main clusters, reflecting their

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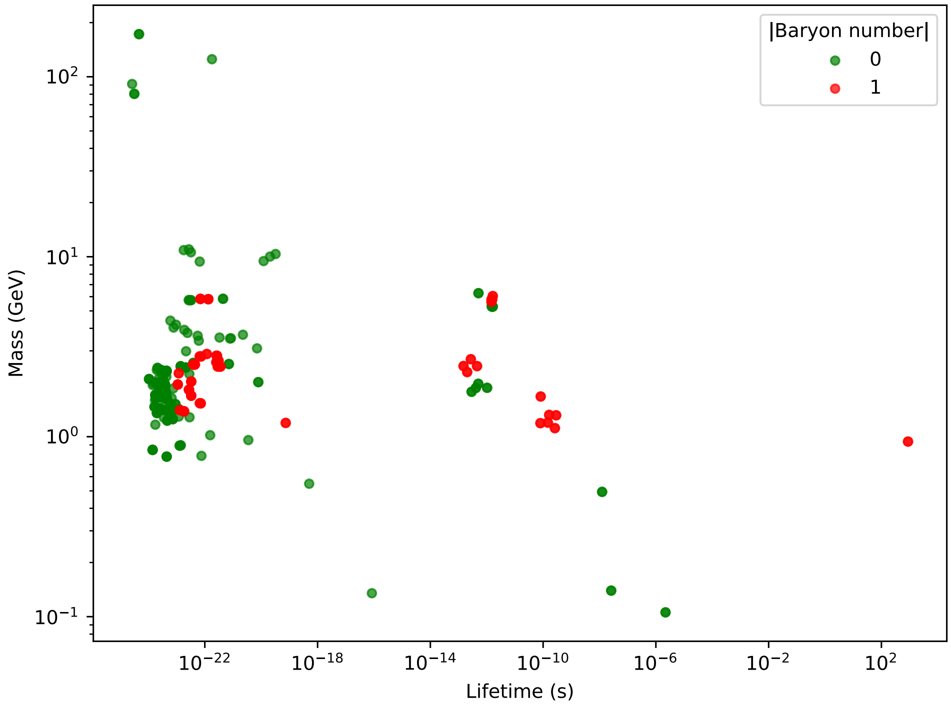

Figure 1. Log-plot of mass versus lifetime for known Standard Model particles in PDG database with measured values. Each point represents a particle, with color indicating its baryon number.

intermediate lifetimes. This is consistent with the fact that both particles primarily decay via electromagnetic interactions. We also observe that baryons and mesons can be distinguished based on their mass. Specifically, all baryons have a mass equal to or greater than that of the proton, while mesons have masses that range between the electron and proton masses. However, this classification criterion no longer holds once the particles discovered later are included. It is therefore interesting to explore whether baryons and mesons can still be classified using alternative methods based solely on experimental observations. This possibility will be examined in the following section.

3.2 Baryon-meson classi cation

The clear clustering discussed in the previous section motivates the use of unsupervised machine learning techniques to uncover more subtle patterns when higher-dimensional information is taken into account. In this and next sections, we build on this idea, showing how such techniques can recover and even extend our understanding of particle classification based on physical properties alone.

In the context of particle physics, each particle decays into various products, such as leptons, neutrinos, or hadrons. These decay products form a set of features that can be used to describe the particle's decay mode. More precisely, the input to the machine can be structured as a dataframe with particles as rows and particles as columns. For each row particle, we examine the decay products across all of its decay modes. The entry at a given row-column position is set to 1 if the column particle appears as a decay product